

BILECIK, Turkey

WMO: 171200

Lat: 40.1502N

Lon: 29.9684E

Elev: 1768

StdP: 13.78

Time Zone: 3.00 (E03)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.6	(c) 15.7	(d) 4.7	(e) 7.5	(f) 13.6	(g) 9.0	(h) 9.3	(i) 18.3	(j) 21.2	(k) 34.1	(l) 18.5	(m) 36.2	(n) 3.3	(o) 40	(p) 0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.0	94.3	66.6	91.2	65.9	88.0	64.7	70.8	86.3	69.0	84.4	67.6	82.2	7.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	102.4	77.9	64.1	95.8	76.4	62.1	89.2	75.3	35.7	86.1	34.2	84.7	32.9	82.5	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.1	(b) 16.3	(c) 13.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 19.1	(b) 16.3	(c) 13.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	32.4	34.9	45.2	51.8	60.3	67.5	74.2	73.4	64.0	56.7	46.4	36.4	
	DBStd	15.70	8.84	7.68	8.15	7.11	5.74	5.89	4.99	5.45	5.74	7.04	7.71	7.17	
	HDD50	1822	546	425	189	58	2	0	0	0	0	24	154	424	
	HDD65	4875	1011	843	614	399	167	38	2	7	84	266	558	886	
	CDD50	3176	1	2	40	113	321	525	749	724	420	231	47	3	
	CDD65	754	0	0	0	4	22	113	286	266	54	8	1	0	
	CDH74	9208	0	0	4	78	316	1441	3372	2980	866	150	1	0	
	CDH80	3769	0	0	0	9	52	498	1580	1346	250	34	0	0	
Wind	WSAvg	5.2	4.0	5.2	5.3	5.5	5.2	6.3	6.7	7.1	5.8	4.4	3.4	3.7	
Precipitation	PrecAvg	17.6	2.0	1.5	1.7	1.6	2.0	1.5	0.8	0.6	0.8	1.5	1.5	2.3	
	PrecMax	24.6	4.7	5.3	3.6	3.0	4.6	5.2	4.2	2.7	2.8	4.4	3.5	6.2	
	PrecMin	12.6	0.4	0.4	0.2	0.3	0.5	0.1	0.0	0.0	0.0	0.1	0.2	0.1	
	PrecStd	3.1	1.3	0.9	0.9	0.7	1.2	1.1	0.9	0.7	0.7	1.0	0.8	1.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.0	60.2	72.0	80.9	84.5	93.7	98.9	96.6	89.7	84.4	71.9	58.5	
		MCWB	48.3	47.8	52.1	58.1	63.3	68.2	67.4	66.2	64.5	64.7	55.7	48.8	
	2%	DB	51.7	55.7	68.2	75.8	80.9	88.8	95.3	93.4	86.1	78.0	67.0	53.7	
		MCWB	44.5	46.1	51.1	57.2	60.6	64.9	66.6	66.9	62.7	58.8	53.7	46.8	
	5%	DB	48.0	52.0	64.4	71.4	77.3	85.9	91.7	91.0	82.7	73.5	63.0	49.6	
		MCWB	42.4	44.4	49.8	54.6	59.5	63.6	65.8	65.9	61.7	57.3	51.7	43.9	
	10%	DB	44.7	48.2	60.2	66.6	73.7	82.4	88.2	87.8	79.2	69.3	59.3	46.5	
		MCWB	41.3	42.4	48.4	52.9	58.1	62.1	64.9	65.0	60.8	56.1	49.8	43.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.2	50.5	54.4	60.8	66.5	71.8	72.8	73.2	68.3	66.5	56.7	49.8
			MCDB	56.4	57.3	68.5	75.3	79.1	88.6	89.7	88.6	83.5	83.6	69.3	57.2
2%		WB	45.0	47.3	52.3	58.0	63.9	69.0	70.3	71.0	65.6	61.6	54.4	47.2	
		MCDB	49.6	53.6	65.6	73.1	75.4	82.7	86.7	86.3	80.4	70.9	64.8	52.5	
5%		WB	43.1	45.0	50.5	55.6	61.6	66.5	68.8	69.1	63.6	59.7	52.7	44.9	
		MCDB	47.3	51.4	61.8	69.0	72.2	79.6	85.1	83.7	78.0	67.8	61.9	48.8	
10%		WB	41.0	42.3	48.8	53.6	59.6	64.3	67.4	67.4	61.9	58.0	50.7	42.7	
		MCDB	44.6	47.5	59.1	65.3	70.5	77.8	82.7	81.6	76.1	65.7	58.1	46.0	
Mean Daily Temperature Range		5% DB	MDBR	13.9	18.7	22.6	22.7	24.2	25.7	27.0	26.2	27.2	21.7	20.9	13.9
			MCDBR	18.7	26.6	30.7	31.0	29.5	30.3	31.4	29.6	32.8	29.9	26.3	18.3
	MCWBR		12.6	17.5	16.3	15.8	13.6	12.5	11.7	11.0	15.3	15.6	15.5	12.2	
	5% WB	MCDBR	16.1	22.7	26.7	26.4	23.6	25.0	25.7	25.7	28.0	22.4	22.8	15.1	
		MCWBR	11.4	15.6	14.6	14.4	12.0	11.3	10.4	10.9	14.0	12.5	13.6	10.5	
Clear-Sky Solar Irradiance	taub		0.326	0.353	0.396	0.436	0.413	0.395	0.395	0.399	0.385	0.370	0.338	0.327	
	taud		2.390	2.314	2.202	2.114	2.248	2.335	2.347	2.330	2.361	2.386	2.432	2.424	
	Ebn at Noon		263	271	270	266	275	279	278	274	271	262	257	253	
	Edh at Noon		27	33	42	49	43	40	39	38	35	31	26	24	
All-Sky Solar Radiation	RadAvg		543	794	1161	1545	1903	2186	2302	2023	1539	1039	714	500	
	RadStd		72	113	110	170	167	201	92	122	124	87	88	72	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-175	N/A	+83

Nomenclature: See separate page